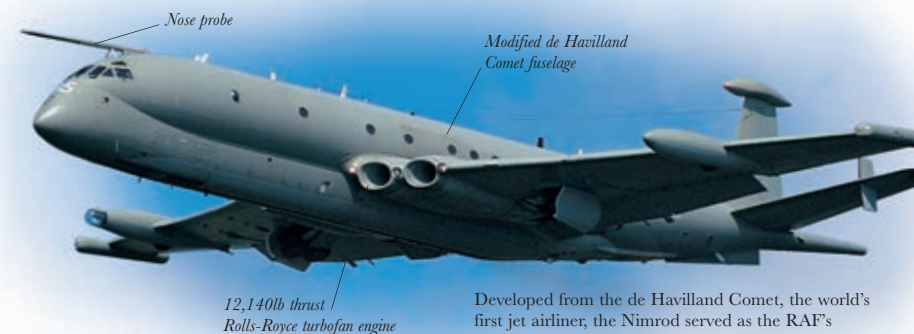


Hawker Siddeley Nimrod MR.1/2



ENGINES 4 x 12,140lb (5,510kg) thrust Rolls-Royce turbofan

WINGSPAN 114ft 10in (35m) **LENGTH** 129ft 1in (39.3m)

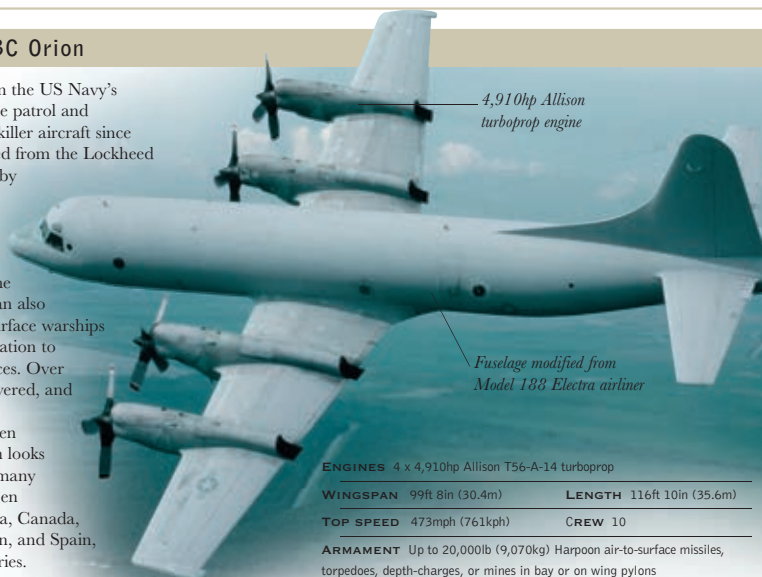
TOP SPEED 576mph (927kph) **CREW** 11

ARMAMENT 13,500lb (6,120kg) Harpoon air-to-surface missiles, torpedoes, depth charges, or mines; 2 x Sidewinder air-to-air missiles

Developed from the de Havilland Comet, the world's first jet airliner, the Nimrod served as the RAF's maritime reconnaissance and antisubmarine aircraft from 1969. A few Nimrods were converted to carry detection equipment only for the Electronic Intelligence (ELINT) role. In the early 21st century, the aircraft were deployed in support of ground operations in Afghanistan and Iraq. The Nimrod was retired from service in 2011.

Lockheed P-3C Orion

The Orion has been the US Navy's land-based maritime patrol and submarine hunter/killer aircraft since 1961. It was adapted from the Lockheed Model 188 Electra by shortening the airframe by 7ft 4in (2.2m) and incorporating a weapons bay in the lower fuselage. It can also be used to attack surface warships and to relay information to friendly surface forces. Over 500 have been delivered, and since a proposed replacement has been canceled, the Orion looks set to continue for many years. It has also been supplied to Australia, Canada, New Zealand, Japan, and Spain, among other countries.



ENGINES 4 x 4,910hp Allison T56-A-14 turboprop

WINGSPAN 99ft 8in (30.4m) **LENGTH** 116ft 10in (35.6m)

TOP SPEED 473mph (761kph) **CREW** 10

ARMAMENT Up to 20,000lb (9,070kg) Harpoon air-to-surface missiles, torpedoes, depth-charges, or mines in bay or on wing pylons

Lockheed U-2A



This high-altitude spy plane was developed in the 1950s and was still in use in the 21st century. From 1956 to 1960, U-2s flew over 30 photographic reconnaissance missions over the former Soviet Union, flying over 14 miles (22km) high. In 1960, a U-2 was shot down and the pilot put on trial in Moscow. In 1962, U-2 photographs revealed Soviet long-range missiles in Cuba.

ENGINE 11,200lb (5,085kg) thrust P&W J75-P-13B turbojet

WINGSPAN 80ft (24.4m) **LENGTH** 49ft 7in (15.6m)

TOP SPEED 494mph (795kph) **CREW** 1

ARMAMENT None

Myasishchev M-55 Geofizika

The Soviet equivalent of the American U-2 high-altitude spy plane was developed in the 1970s, and its designers claimed their aircraft was far more stable. The original M-17 was developed into the M-55 Geofizika, which the Russians claim is used for atmospheric monitoring. The Americans, of course, claim the same for most U-2 flights.



ENGINES 2 x 11,025lb (5,000kg) thrust Aviadvigatel turbojet

WINGSPAN 122ft 11in (37.4m) **LENGTH** 75ft (22.8m)

TOP SPEED 466mph (750kph) **CREW** 1

ARMAMENT None

straining of muscles, which can help them withstand much higher gs for short periods.

Pilots also need equipment, up to and including full-pressure space suits, for flights at extreme altitude. The design of flight helmets has improved dramatically since the first one was created. They were once heavy enough to cause frequent neck injuries, but are now lightweight while offering the same protection. All modern fighter pilots fly in masks, which, as well as acting as an intercom, provide the oxygen supply. Pilots have to learn to regulate the oxygen appropriately and recognize any sign of hypoxia – oxygen starvation of the brain – almost instantly.

Flying a modern single-seat fighter not only imposes physical strains but also makes intense mental demands on pilots. A combination of complex multimode radars, infrared navigation and targeting equipment, sensors to identify hostile radars or missiles, and data links to AWACS (airborne warning and control system) aircraft can provide more information input than a human brain can readily cope with. At the same time as processing this flow of information, the pilot has to operate his sophisticated targeting and weapons systems for attack, and deploy his defensive array. He also, of course, has to fly the airplane.

Improvements in computing have made it possible to organize and analyze the flow of data from different sources so that the pilot is presented with the most relevant information in increasingly easy-to-use graphic form. There is a move toward allowing computers or airborne controllers to make decisions about targeting and weapon launch independently of the pilot. Indeed, with unmanned, remotely piloted vehicles (RPVs) ever more common in combat use, there have been hints that fighter pilots might almost have had their day.

The Gulf War

Most of the general public became aware of the modern revolution in air warfare in 1991 when Operation Desert Storm unleashed the aerial might of the United States and its Coalition allies against Iraq. The conflict displayed, among many other Coalition strengths, the effectiveness of the direction of the battle by airborne controllers, the usefulness of “stealth” technology, and the progress that had been made with missile guidance and smart bombs.

During the Gulf War, four Boeing E-3 Sentry AWACS aircraft were in the air at all times, one for each of the sectors into which the battle area had been divided, and one spare for backup. The E-3 is a Boeing 707 airliner adapted to carry radars of great power and discrimination. Inside the aircraft, rows of display consoles were manned by highly trained personnel, fighting to